Mohammad Aziz

Full Stack Engineer

Lahore, Pakistan | (+92) 321-6006717 | mohammadazizramay@gmail.com
mohammadaziz.me | github.com/mohdzez | linkedin.com/in/mohdzez

Full stack engineer with 5+ years building products end to end, and founding engineer at two startups. Most recent work is LLM systems (MCP servers, agent pipelines, evals) shipped inside real products, plus production WhatsApp messaging on Meta's Cloud API. Builds with Claude Code daily and ships developer tools on the side (Liftoff).

WORK EXPERIENCE

Venturenox - Full Stack Engineer, AI Systems and Backend Infrastructure

Feb 2025 - Present | Lahore, Pakistan | Product: SarahAI, an AI executive assistant for SMEs on WhatsApp, voice and mobile

- Built the MCP server that routes model calls to fallback OSS models (120B and 20B) on Groq and Cerebras by live latency and cost, which removed the single-provider dependency. Also built CRM integrations over MCP across HubSpot, Pipedrive, Zoho and Odoo.
- Wired a multi-model pipeline across OpenAI, Gemini and Claude with ElevenLabs for voice, with prompts versioned and traced in Langfuse to catch quality regressions.
- Moved job routines onto Inngest as durable multi-step flows, with agent steps running inside them, and replaced Celery CRON scheduling with AWS Lambda, SQS and EventBridge. Time-based WhatsApp delivery now scales without anyone tuning workers.
- Migrated the messaging layer from Respond.io to Meta's native WhatsApp Cloud API as a live hot swap. Users mid-conversation saw no downtime, and the move opened up read receipts, typing indicators, GIFs and the other native message types Respond.io had been holding back.
- Built the notification system around opt-in consent templates and automated renewal flows, with no Meta policy violations to date, and added OneSignal push alongside it.
- Own subscription lifecycle and expiry-based feature gating, and added a cache layer across the API-heavy paths that cut redundant calls and response times.

Stack: Django, Python, PostgreSQL, AWS Lambda/SQS/EventBridge, Inngest, MCP, RAG, Langfuse, OpenAI, Gemini, Claude, ElevenLabs, Groq, Cerebras, OneSignal, WhatsApp Cloud API

Informly - Full Stack Engineer (Founding)

Jan 2023 - Feb 2025 | Lahore, Pakistan | Business growth platform, informly.co

- First engineer. Made the architecture calls across the whole stack and took the product from an empty repo to paying enterprise customers.
- Built the AI Engine: graph-based tenant memory, an eval suite with LLM-as-judge scoring, and per-step visibility into agent runs to trace a bad answer to its cause.
- Built a drag-and-drop workflow builder on Hatchet that let non-technical operators automate multi-step work themselves, replacing most of their manual spreadsheets.
- Moved the API layer from REST to GraphQL on Apollo and added in-memory caching. Over-fetching went away and page loads got noticeably faster.
- Shipped multi-location accounts with fine-grained RBAC tiers for enterprise clients, plus customer ticketing and a notification layer each client could configure.
- Automated lead capture and follow-ups with n8n pipelines and Supabase, plus a lightweight in-house CRM.

Stack: Next.js, TypeScript, Node.js, Nx monorepo, Apollo GraphQL, Nexus, PostgreSQL, Hatchet, Python, Pydantic, OpenAI, Anthropic, n8n, Supabase

Alharam Hostel - Full Stack Engineer

Jan 2022 - Feb 2023 | Lahore, Pakistan

- Built a biometric attendance system from scratch on ZKTeco hardware (Python SDK), using WebSockets and SSE so scans landed in the dashboard live. Added D3.js and Recharts analytics with alerts for late clock-ins and unrecognized scans.

RipeOnTime - Software Engineer, then Lead Engineer

Jul 2021 - May 2022 | Remote (Texas, USA)

- Founding engineer on a grocery and vendor marketplace. Owned the admin portal from first design to production: tiered subscriptions and vendor, admin and super-admin roles across storefronts.
- Integrated Stripe billing with webhooks for renewals, failures and cancellations; removed manual release steps from the Azure DevOps pipeline. Promoted to lead: ran code reviews, set standards, mentored juniors and helped hire.

System Junctions - Frontend Developer (Intern, then Junior)

Dec 2020 - Jan 2022 | Lahore, Pakistan

- Led the frontend for MediCorp, a hospital group's clinical platform, and built shared Storybook component libraries in React, React Native and Angular.

PROJECTS

Liftoff - Built solo | tryliftoff.tech | Push code. Get infrastructure. Keep both.

- A deploy platform that provisions into the user's own cloud. Services drawn on a visual canvas become the deploy spec; a git push builds images with GitHub Actions and Pulumi provisions them on DigitalOcean or Azure (beta), including Kubernetes, so cloud credentials never pass through Liftoff.
- Per-PR previews, gated migrations, per-service rollbacks, live logs, a CLI and REST API, and pre-deploy cost estimates. Provisioning runs through a BullMQ queue pipeline.

SKILLS

Languages and data: TypeScript, JavaScript, Python, SQL, PostgreSQL, MongoDB, Neo4j, FalkorDB, Prisma, TypeORM

Frontend: React, Next.js, Redux, Tailwind CSS, Storybook, D3.js, Recharts, Ant Design, Material UI

Backend and APIs: Node.js, Nest.js, Express, Django, GraphQL, Apollo, REST, webhooks, Inngest, Hatchet, Celery, BullMQ, WhatsApp Cloud API, Stripe

AI and LLMs: Claude Code, MCP, RAG, agent orchestration, Langfuse, evals, LLM-as-judge, Graphiti, OpenAI, Anthropic, Gemini, Groq, Cerebras, ElevenLabs

Cloud and DevOps: AWS (Lambda, SQS, EventBridge), GCP, DigitalOcean, Azure, Docker, Nixpacks, Kubernetes (DOKS, AKS), Pulumi, GitHub Actions, Azure DevOps

EDUCATION

University of Central Punjab, Lahore, Pakistan - B.Sc. Computer Science